Section 10 bootstrap, with the two open gaps marked

Benjamin Stanley Frohman

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Assume a smooth divergence-free solution of unforced 3D incompressible Navier–Stokes on $\mathbb{T}^3$ exists on $[0,T)$. Let $M(t) = \|\omega(t)\|_{L^\infty}$ and $M_\varepsilon(t) = \|\omega * \eta_\varepsilon\|_{L^\infty}$.

Auxiliary field.

$$\partial_t \phi + u \cdot \nabla \phi = |\omega|^2, \qquad \phi(0,x) = 0.$$

Along characteristics, $\|\phi(t)\|_{L^\infty} \le \int_0^t M(s)^2\, ds$. For constant $M(s) = M_0 > 0$, the right-hand side equals $M_0^2 t$; more generally a positive lower bound on M makes that right-hand side grow at least linearly. This does not itself give a lower bound on ϕ, and it is not a uniform constant on $[0,\infty)$.

Pointwise stretching. At an interior maximum of $|\omega_\varepsilon|^2$ the viscous pairing is non-positive. The pointwise algebraic stretching estimate is

$$|\omega_\varepsilon \cdot ((\omega_\varepsilon \cdot \nabla)u)| \le M_\varepsilon(t)^2\, \|\nabla u(t)\|_{L^\infty}.$$

The strain factor $\|\nabla u(t)\|_{L^\infty}$ is essential and is not a universal constant. Calderón–Zygmund theory gives BMO rather than a closed L^∞ estimate from bounded vorticity. A logarithmic Beale–Kato–Majda estimate also involves a higher-regularity norm; it does not justify dropping the strain factor. A PDE evolution estimate for mollified vorticity additionally requires control of the mollification commutators.

Lyapunov inequality, as stated.

$$\frac{dS_\varepsilon}{dt} \le C_{\mathrm{abs}}\big(1 + M_\varepsilon(t)^3 \|\phi_\varepsilon\|_{L^\infty}^{3/2}\big) - \kappa' \int_{\mathbb{T}^3} |\omega_\varepsilon|^6\, d\lambda.$$

Gap A. For constant positive M_ε, substituting the history bound introduces a factor proportional to $t^{3/2}$ rather than a time-independent constant. In general this substitution alone does not justify the autonomous comparison ODE $y' = Cy^2 - \kappa'' y^3$.

Gap B. The relation $\int |\omega_\varepsilon|^6 \geq c M_\varepsilon^4$ does not follow from Hölder. Hölder supplies $\|\omega_\varepsilon\|_6 \leq M_\varepsilon$ on a unit-volume torus. A lower bound requires a lower bound on the measure of $\{|\omega_\varepsilon| \gtrsim M_\varepsilon\}$.

A conditional analytic sufficient condition. For the unmollified smooth solution, put $q = |\omega|^2$ and $S = (\nabla u + (\nabla u)^T)/2$. The vorticity equation gives

$$(\partial_t + u \cdot \nabla - \nu\Delta)q = 2\omega \cdot S\omega - 2\nu|\nabla\omega|^2.$$

Set $a(t) = \sup_x \max\{\omega \cdot S\omega/|\omega|^2, 0\}$, taking the quotient to be zero where $\omega = 0$. If the additional hypothesis $A = \int_0^T a(t)\,dt < \infty$ holds, the maximum principle implies

$$M(t) \leq M(0) \exp\left(\int_0^t a(s)\,ds\right), \qquad \int_0^T M(t)\,dt \leq T M(0) e^A < \infty.$$

This is a conditional classical analytic implication, not a Lean-formalized PDE estimate here. The solution-derived bound $A < \infty$ is not supplied by the tether or by the integer Riccati rearrangement and remains an open input.

What would close (B). A proof must produce $\int_0^T M(t)\,dt < \infty$ for every finite T, for every smooth divergence-free finite-energy u°, with $f \equiv 0$, and then invoke Beale–Kato–Majda plus parabolic regularity. Neither the tether degeneracy identity $B(F, H) \equiv 0$ nor a holographic analogy supplies that integral bound.